

Sector-Resolved Critical-Branch Returns at a Quadratic Band-Merging Map:

Exact Two-Channel Factorization, a Dark Antisymmetric Mode,
and a Conditional Cubic-Phase Law

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July 2026

Abstract

At the first algebraic band-merging parameter of $f_u(x) = 1 - ux^2$, the positive Gaussian return through both critical preimages has a principal radius near 0.79, whereas a single branch and the parity-extracted physical bulk edge lie near 0.756 at the smallest computed noise. The omitted branch is therefore not a small complement. This paper asks whether a branch phase can supply the missing finite-channel structure without confusing a time lift with the physical spectrum.

Let $\mathcal{C}$ carry the endpoint packet into the direct sum of the left and right critical windows, let $\mathcal{E}$ return that branch space to the endpoint, and put $J_\theta = \text{diag}(1, e^{i\theta})$. We prove the exact factorization

$$\mathcal{R}(\theta) = \mathcal{E}J_\theta\mathcal{C} = \mathcal{R}_- + e^{i\theta}\mathcal{R}_+, \quad \mathcal{B}(\theta) = J_\theta\mathcal{C}\mathcal{E},$$

and the Sylvester identity

$$\det(\mathbf{I} - z\mathcal{R}(\theta)) = \det(\mathbf{I} - z\mathcal{B}(\theta)).$$

Thus endpoint and branch formulations have the same nonzero spectrum; the two-channel reduction is not a second spectral ansatz.

If the branch monodromy closes on one profile per branch and has rank one, its only nonzero eigenvalue is $\eta_- + e^{i\theta}\eta_+$. Exact branch equivalence makes the unweighted channel bright on the symmetric vector and dark on the antisymmetric vector. Among unit-modulus relative weights, the requirement that the combined return retain one-branch modulus then forces $\theta = \pm 2\pi/3$. This cubic-phase law is conditional. We also prove that radial data alone cannot identify it: the half-weighted positive sum $(\mathcal{R}_- + \mathcal{R}_+)/2$ has the same leading modulus in the symmetric rank-one limit. Moreover, a discrete period- k Floquet sector contains the cubic phase only when 3 divides k .

Seven-noise computations show that a two-profile compression recovers the full two-branch bright return while its dark/bright eigenvalue ratio falls to 1.45×10^{-5} at $\sigma = 10^{-4}$. At that noise, the phase preserving one-branch modulus is 0.665704π , the phase matching the archived bulk edge is 0.661273π , and the fixed cubic-phase radius is 0.756336 versus the bulk value 0.757023. These are floating-point diagnostics, not a derivation of the physical phase. At $\sigma = 10^{-3}$, inserting Perron/parity deflation before localization changes the entire phase family, confirming that phase selection and peripheral extraction remain coupled. The next valid target is therefore a phase-dependent, peripherally biorthogonal Grushin problem that derives its branch weights rather than fitting them.

Keywords: Gaussian transfer operator; critical branch; Sylvester determinant identity; rank-one monodromy; Floquet phase; Grushin problem; non-normal spectrum; quadratic map.

MSC 2020: 37E05; 37D25; 47A10; 47A55; 47B65; 60J05; 65P30.

1 Introduction

The small-noise spectrum of a row-normalized Gaussian perturbation of a quadratic band-merging map has a deterministic pole, a half-logarithmic resolution clock, and a time-ordered boundary cycle [8, 9, 11, 14]. A branch-isolated Gaussian realization of that cycle produces compact local returns whose principal radius accurately follows the outer parity-extracted bulk edge at the smallest archived noises [12]. It also produces an exact roots-of-unity ring after a cyclic time lift.

The complement test in the preceding paper found two obstructions to turning that observation directly into a physical spectral theorem [10]. First, the endpoint Gaussian pulls back through two critical preimages with asymptotically equal local L^2 mass. The second branch is not a far Gaussian tail. Numerically, the return through both branches is already almost the unrestricted positive endpoint return. Second, a sector-free time lift is an exact direct sum of Floquet rotations of the physical two-step operator. A roots-of-unity ring in that lift does not count distinct physical resonances.

Those negative statements do not close the route. They replace a scalar packet with a two-channel branch space and require the closing phase to be tracked explicitly. The present paper studies this smallest corrected object. Three questions are separated:

- (i) What is the exact relation between a phase-weighted endpoint return and a return acting on the two critical branches?
- (ii) What does a one-packet rank-one limit imply about bright and dark branch combinations?
- (iii) Can the phase that repairs the factor-of-two bright return be inferred from radial agreement alone?

The answer to the first question is an exact Sylvester factorization. The answer to the second is a rank-one bright/dark law. The third answer is deliberately mixed: unit-modulus weighting uniquely selects a cubic phase in the symmetric limit, and the cleanest numerical endpoint lies close to it, but a real half-weight normalization gives the same radial outcome. The dynamics has not yet selected between them.

Main results and logical status

- S1. Exact two-channel factorization.** The endpoint return is $\mathcal{E}J_\theta\mathcal{C}$ and the branch return is $J_\theta\mathcal{C}\mathcal{E}$. Their nonzero spectra, algebraic multiplicities, trace powers, and Fredholm determinants agree exactly.
- S2. Rank-one bright/dark algebra.** A one-profile-per-branch rank-one monodromy has one nonzero eigenvalue $\eta_- + e^{i\theta}\eta_+$. Under branch exchange symmetry, the unweighted channel is symmetric and its antisymmetric channel is exactly dark. This statement is an exact consequence of the rank-one reduction; operator-norm convergence of the full Gaussian return to that reduction is not proved here.
- S3. Conditional cubic-phase theorem.** For positive branch amplitudes, the phase producing any admissible target modulus is explicit. Equal branch amplitudes and preservation of one branch force $\theta = \pm 2\pi/3$ when both branch weights have modulus one.
- S4. Normalization non-identifiability.** In the same symmetric rank-one limit, the positive half sum and the unit-weight cubic-phase sum have identical leading modulus. Radius data alone cannot decide whether the missing ingredient is a closing phase or a dual entrance/exit normalization.
- S5. Discrete-sector obstruction.** The ordinary period- k time sectors contain $e^{\pm 2\pi i/3}$ if and only if $3 \mid k$. Since the intrinsic clock takes general integer values, a universal cubic law cannot be supplied by choosing a fixed discrete time sector at every noise.

S6. Floating-point branch audit. A two-profile compression resolves the positive two-branch return at all seven archived scales. Its smaller eigenvalue remains at most 4.1% of the bright eigenvalue and reaches 1.45×10^{-5} at the smallest noise. Phase agreement with $2\pi/3$ is strongest at the cleanest tail point, not uniform across the staircase clock. No interval, pseudospectral, or asymptotic enclosure is claimed.

The logical outcome is

$$\begin{array}{l} \text{two critical branches} \implies \text{exact phase-dependent branch determinant,} \\ \text{rank-one closure + unit branch weights} \implies \theta = \pm 2\pi/3, \\ \text{radial numerical agreement} \not\Rightarrow \text{physical cubic phase,} \\ \text{derived Grushin weights + complement control} \implies ? \text{ physical bulk determinant.} \end{array} \quad (1)$$

The first two implications and the non-implication are proved here. The last arrow is the next operator problem.

2 Gaussian map and branch-resolved return

Let u_c be the root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (2)$$

and define

$$f(x) = 1 - u_c x^2, \quad \lambda = 2u_c(u_c - 1). \quad (3)$$

Numerically,

$$u_c = 1.543689012692076 \dots, \quad \lambda = 1.678573510428322 \dots \quad (4)$$

The physical partition separating the two critical preimages is

$$b = u_c^{-1/2}. \quad (5)$$

For $\sigma > 0$, let $\mathcal{K}_\sigma$ be the backward Markov operator on $\mathcal{H} = L^2(0, 1)$ with row-normalized folded Gaussian kernel

$$P_\sigma(x, y) = \frac{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))}{Z_\sigma(x)}, \quad (6)$$

$$Z_\sigma(x) = \int_0^1 \{\phi_\sigma(y - f(x)) + \phi_\sigma(y + f(x))\} dy, \quad (7)$$

$$\phi_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} e^{-t^2/(2\sigma^2)}. \quad (8)$$

Write

$$\mathcal{T}_\sigma = \mathcal{K}_\sigma^2. \quad (9)$$

At fixed noise, $\mathcal{K}_\sigma$ is Hilbert–Schmidt and $\mathcal{T}_\sigma$ is trace class. All products below are therefore covered by the standard trace-ideal cyclicity and Fredholm determinant identities [1, 5].

Let $P_0, \dots, P_{k-2}$ be the time-labeled packet projections along the boundary cycle. The final critical slice is split by

$$P_{k-1} = P_- + P_+, \quad P_- P_+ = 0, \quad (10)$$

where P_- lies to the left of b and P_+ is its critical sibling. Put

$$\mathcal{H}_0 = \text{ran } P_0, \quad \mathcal{H}_{\text{br}} = \text{ran } P_- \oplus \text{ran } P_+. \quad (11)$$

The packet geometry and the critical-profile scaling used to choose these windows are those of RH-18 and RH-19. The algebra below only requires the orthogonal splitting in (10).

Define the critical entrance map $\mathcal{C} : \mathcal{H}_0 \rightarrow \mathcal{H}_{\text{br}}$ by

$$\mathcal{C}v = \begin{pmatrix} P_- \mathcal{T}_\sigma P_0 v \\ P_+ \mathcal{T}_\sigma P_0 v \end{pmatrix}, \quad (12)$$

and the return-to-endpoint map $\mathcal{E} : \mathcal{H}_{\text{br}} \rightarrow \mathcal{H}_0$ by

$$\mathcal{E}(w_-, w_+) = P_0 \mathcal{T}_\sigma P_1 \mathcal{T}_\sigma \cdots P_{k-2} \mathcal{T}_\sigma (w_- + w_+). \quad (13)$$

Thus $\mathcal{C}$ performs the endpoint-to-critical pullback, while $\mathcal{E}$ performs the remaining $k-1$ two-step channels. Introduce

$$J_\theta = \begin{pmatrix} \mathbf{I} & 0 \\ 0 & e^{i\theta} \mathbf{I} \end{pmatrix} \quad \text{on } \mathcal{H}_{\text{br}}. \quad (14)$$

Definition 2.1 (Endpoint and branch returns). The phase-weighted endpoint return and branch monodromy are

$$\mathcal{R}(\theta) = \mathcal{E} J_\theta \mathcal{C} \quad \text{on } \mathcal{H}_0, \quad \mathcal{B}(\theta) = J_\theta \mathcal{C} \mathcal{E} \quad \text{on } \mathcal{H}_{\text{br}}. \quad (15)$$

If $\iota_\pm$ and $\pi_\pm$ denote the canonical injections and projections of the branch sum, define

$$\mathcal{R}_- = \mathcal{E} \iota_- \pi_- \mathcal{C}, \quad \mathcal{R}_+ = \mathcal{E} \iota_+ \pi_+ \mathcal{C}. \quad (16)$$

Then linearity gives the exact phase family

$$\boxed{\mathcal{R}(\theta) = \mathcal{R}_- + e^{i\theta} \mathcal{R}_+}. \quad (17)$$

At $\theta = 0$, this is precisely the branch-complete positive return of RH-19. At other phases it is a complex auxiliary return and is no longer a Markov-positive operator.

3 Exact endpoint–branch determinant identity

The branch space is not introduced as an independent spectral model. It is the reversed product associated with the same entrance and exit maps.

Theorem 3.1 (Two-channel Sylvester identity). *For every $\theta \in \mathbb{R}$, the nonzero spectra of $\mathcal{R}(\theta)$ and $\mathcal{B}(\theta)$ agree, including algebraic multiplicity. Their trace powers satisfy*

$$\text{tr } \mathcal{R}(\theta)^m = \text{tr } \mathcal{B}(\theta)^m, \quad m \geq 1, \quad (18)$$

and their Fredholm determinants obey

$$\boxed{\det_{\mathcal{H}_0}(\mathbf{I} - z\mathcal{R}(\theta)) = \det_{\mathcal{H}_{\text{br}}}(\mathbf{I} - z\mathcal{B}(\theta))} \quad (z \in \mathbb{C}). \quad (19)$$

The same statements hold as exact matrix identities for every finite Nyström discretization.

Proof. Set $A = \mathcal{E}$ and $B = J_\theta \mathcal{C}$. Then $AB = \mathcal{R}(\theta)$ and $BA = \mathcal{B}(\theta)$. The standard AB – BA theorem identifies their nonzero eigenvalues and algebraic multiplicities [4]. Cyclicity of the trace gives

$$\text{tr}(AB)^m = \text{tr}(BA)^m.$$

Finally, the Sylvester–Fredholm identity $\det(\mathbf{I} - zAB) = \det(\mathbf{I} - zBA)$ proves (19) [1, 5]. $\square$

Corollary 3.2 (No spectral cost for moving to branch space). *Any nonzero zero of the endpoint determinant on the left side of (19) is a zero of the branch determinant on the right side with the same order, and conversely. Consequently a branch-space effective Hamiltonian may be used without changing the finite-return spectrum, provided that the full branch spaces and exact maps $\mathcal{C}, \mathcal{E}$ are retained.*

The final qualification matters. Replacing each infinite-dimensional branch by one profile is a Galerkin reduction and does require an error estimate. The determinant identity isolates that approximation from the exact branch reorganization.

4 Rank-one closure and the dark channel

We next state the finite-channel algebra that would follow if one normalized profile on each critical lobe captures the leading return. It is useful to separate this exact conditional algebra from the still-unproved asymptotic closure.

Identify the two profile amplitudes with $\mathbb{C}^2$. For a column $c = (c_-, c_+)^T$ and a row functional $\ell = (\ell_-, \ell_+)$, write

$$(c \otimes \ell)v = c\ell(v). \quad (20)$$

Put

$$\eta_- = \ell_- c_-, \quad \eta_+ = \ell_+ c_+. \quad (21)$$

Proposition 4.1 (Rank-one phase law). *Suppose the unweighted two-profile branch monodromy is*

$$B_0 = c \otimes \ell. \quad (22)$$

Then

$$B_\theta = J_\theta B_0 = (J_\theta c) \otimes \ell \quad (23)$$

has rank at most one, and its only possible nonzero eigenvalue is

$$\boxed{\Lambda(\theta) = \eta_- + e^{i\theta} \eta_+}. \quad (24)$$

Equivalently,

$$\det(\mathbf{I} - zB_\theta) = 1 - z\{\eta_- + e^{i\theta} \eta_+\}. \quad (25)$$

Proof. A rank-one operator $u \otimes \ell$ has possible nonzero eigenvalue $\ell(u)$. Taking $u = J_\theta c$ gives

$$\ell(J_\theta c) = \ell_- c_- + e^{i\theta} \ell_+ c_+.$$

The determinant formula follows from the rank-one determinant lemma. $\square$

Let

$$e_b = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad e_d = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad U = (e_b, e_d). \quad (26)$$

Corollary 4.2 (Symmetric bright mode and antisymmetric dark mode). *If the two branches are exactly equivalent in the profile model, so that $c_- = c_+ = c_0$ and $\ell_- = \ell_+ = \ell_0$, then with $\eta = \ell_0 c_0$,*

$$B_0 = \eta \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad U^* B_0 U = \begin{pmatrix} 2\eta & 0 \\ 0 & 0 \end{pmatrix}. \quad (27)$$

Thus e_b is the bright eigenvector and e_d is an exactly dark antisymmetric eigenvector. The twisted nonzero eigenvalue is

$$\Lambda(\theta) = \eta(1 + e^{i\theta}). \quad (28)$$

At $\theta = \pi$, the return has zero spectral radius in the exact rank-one model.

The last statement is spectral. Without the stronger equality $\mathcal{R}_- = \mathcal{R}_+$, the phase- π operator can be nonzero and nilpotent in an ideal rank-one reduction. The distinction is relevant for non-normal returns [6].

Definition 4.3 (Two-profile compression). Let ψ_- and ψ_+ be normalized critical pullback profiles supported on the two branch windows. The computed branch matrix is

$$B_{ij}^{(2)} = \langle \psi_i, \mathcal{B}(0)\psi_j \rangle, \quad i, j \in \{-, +\}. \quad (29)$$

Its larger and smaller eigenvalues are denoted Λ_b and Λ_d by decreasing modulus.

No theorem in this paper asserts $\|\mathcal{B}(0) - c \otimes \ell\| \rightarrow 0$ in the full branch space. The ratio $|\Lambda_d/\Lambda_b|$ and the singular values of the full endpoint return are numerical diagnostics of that missing rank-one estimate.

5 The conditional cubic-phase law

The rank-one formula reduces phase selection to elementary but informative geometry.

Theorem 5.1 (Forced relative phase). *Let $a, b > 0$, and let $\tau \geq 0$ satisfy the triangle inequalities*

$$|a - b| \leq \tau \leq a + b. \quad (30)$$

There is a unique $\theta \in [0, \pi]$ such that

$$|a + e^{i\theta}b| = \tau, \quad (31)$$

and it is given by

$$\boxed{\theta = \arccos\left(\frac{\tau^2 - a^2 - b^2}{2ab}\right)}. \quad (32)$$

In particular, preserving the first-branch modulus $\tau = a$ is possible if $b \leq 2a$ and gives

$$\cos \theta = -\frac{b}{2a}. \quad (33)$$

If $a = b$, then

$$\boxed{\theta = \frac{2\pi}{3}} \quad (34)$$

on $[0, \pi]$, or equivalently $\theta = \pm 2\pi/3$ modulo 2π after complex conjugation.

Proof. Squaring (31) gives

$$\tau^2 = a^2 + b^2 + 2ab \cos \theta.$$

Condition (30) is exactly the condition that the resulting cosine lie in $[-1, 1]$. Cosine is one-to-one on $[0, \pi]$, which proves uniqueness and (32). Setting $\tau = a$ gives (33); setting also $a = b$ gives $\cos \theta = -1/2$. $\square$

Corollary 5.2 (Stability near equal branches). *If $b/a = 1 + \varepsilon$ and the first-branch modulus is preserved, then*

$$\theta = \frac{2\pi}{3} + \frac{\varepsilon}{\sqrt{3}} + O(\varepsilon^2). \quad (35)$$

Proof. Expand $\arccos[-(1 + \varepsilon)/2]$ at $\varepsilon = 0$. $\square$

For a k -component two-step return, the associated one-step radius in the rank-one model is

$$\rho(\theta) = |\eta_- + e^{i\theta}\eta_+|^{1/(2k)}. \quad (36)$$

The exponent $1/(2k)$ converts a k -fold return of $\mathcal{T}_\sigma = \mathcal{K}_\sigma^2$ back to one Markov step.

5.1 Why the theorem is conditional

Theorem 5.1 does not say that the Gaussian dynamics supplies unit-modulus branch coefficients. It says what follows if that Floquet-type normalization is imposed. There is an exact competing normalization.

Proposition 5.3 (Radial non-identifiability). *Suppose $\eta_- = \eta_+ = \eta > 0$ in the rank-one model. Both weighted sums*

$$\frac{1}{2}\mathcal{R}_- + \frac{1}{2}\mathcal{R}_+, \quad (37)$$

$$\mathcal{R}_- + e^{2\pi i/3}\mathcal{R}_+ \quad (38)$$

have leading return eigenvalue of modulus η . Consequently the radial condition “two branches reproduce one-branch modulus” cannot distinguish a real half weight from a unit-modulus cubic phase.

Proof. The scalar factors are $\frac{1}{2}\eta + \frac{1}{2}\eta = \eta$ and $\eta(1 + e^{2\pi i/3})$, whose moduli both equal η . $\square$

The half weight could arise from a dual Grushin normalization or from determinant combinatorics; the cubic phase could arise from a spectral-parameter closing condition. Neither mechanism has yet been derived. Inserting either coefficient by hand would merely fit the desired radius.

6 Why a fixed discrete time sector is insufficient

RH-19 proved that the unrestricted period- k time lift decomposes into the sectors

$$q_{\ell,k} = e^{2\pi i\ell/k}, \quad 0 \leq \ell < k. \quad (39)$$

If the relative branch phase were identified with one of these values, the cubic phase would be arithmetically restricted.

Proposition 6.1 (Cubic phase in a discrete clock). *There exists an integer ℓ for which*

$$e^{2\pi i\ell/k} = e^{\pm 2\pi i/3} \quad (40)$$

if and only if 3 divides k .

Proof. Equation (40) is equivalent to $\ell/k \equiv \pm 1/3 \pmod{1}$. This congruence has an integer solution ℓ exactly when $k/3$ is an integer. $\square$

The seven archived intrinsic periods are

$$k = 3, 4, 5, 6, 7, 8. \quad (41)$$

Only the periods 3 and 6 contain an exact cubic sector. In particular, the close cubic-phase match at the smallest noise occurs at $k = 8$, where no such discrete sector exists. This observation rules out one tempting interpretation: the inferred phase is not simply the label of a fixed ordinary time-Fourier block.

A phase can still enter through a continuous spectral parameter. In a Grushin formulation one expects an effective determinant of the form

$$D_{-+}(z) = \det\{I - z\mathcal{E}J_{\theta(z)}\mathcal{C} - \Sigma(z)\}, \quad (42)$$

where the closing multiplier $e^{i\theta(z)}$ is derived from the physical boundary condition and $\Sigma(z)$ is the complement self-energy. Equation (42) is a target architecture, not a theorem of the present paper.

7 Peripheral extraction remains inside the phase problem

Let E_0 and E_- be the Perron and parity spectral projectors of $\mathcal{K}_\sigma$, with eigenvalues 1 and $\lambda_-(\sigma)$, and set

$$K_b = \mathcal{K}_\sigma - E_0 - \lambda_- E_- . \quad (43)$$

Since the three spectral pieces commute and annihilate one another,

$$K_b^2 = \mathcal{T}_\sigma - E_0 - \lambda_-^2 E_- . \quad (44)$$

For every pair of packet windows,

$$P_j K_b^2 P_{j+1} = P_j \mathcal{T}_\sigma P_{j+1} - P_j E_0 P_{j+1} - \lambda_-^2 P_j E_- P_{j+1} . \quad (45)$$

The correction has rank at most two on each edge but need not have small norm near an endpoint boundary layer [10].

Constructing $\mathcal{C}_b$ and $\mathcal{E}_b$ from K_b^2 preserves the exact phase algebra,

$$\mathcal{R}_b(\theta) = \mathcal{E}_b J_\theta \mathcal{C}_b = \mathcal{R}_{b,-} + e^{i\theta} \mathcal{R}_{b,+} . \quad (46)$$

It does not make $\mathcal{R}_b(\theta)$ a posteriori deflation of the raw $\mathcal{R}(\theta)$. Expanding the k -fold product of (45) produces peripheral insertion terms at every time slice. Their coefficients interact with which branch is selected at the critical slice. Thus phase weighting and peripheral extraction must be implemented in the same entrance/exit construction.

8 Seven-noise two-branch audit

The analytic identities above require no numerical hypothesis. The computation tests whether one profile per branch is a useful finite-channel description and whether the conditional cubic phase is compatible with the archived tail data.

8.1 Protocol

We use the row-normalized folded midpoint matrices of RH-15–RH-19 with

$$n\sigma = 20.48 \quad (47)$$

and an eight-standard-deviation sparse-kernel cutoff. The period is $k = N_\sigma^H + 1$, where N_σ^H is the independently computed Hellinger half-energy rank from RH-16. Every packet mask uses the fixed six-width window.

Let g_0 be the endpoint Gaussian profile. After one two-step pullback, the two normalized lobe profiles are

$$\psi_\pm = \frac{P_\pm \mathcal{T}_\sigma g_0}{\|P_\pm \mathcal{T}_\sigma g_0\|_2} . \quad (48)$$

The full cycle is applied matrix-free to each profile and projected back onto $\text{span}\{\psi_-, \psi_+\}$, giving $B^{(2)}$ in (29). The corresponding one-step bright radius is $|\Lambda_b|^{1/(2k)}$.

The scalar phase diagnostic imports the left and right principal return eigenvalues $\eta_\pm$ from RH-19. It computes

$$\theta_{\text{one}} = \arccos\left(-\frac{\eta_+}{2\eta_-}\right) , \quad (49)$$

$$\theta_{\text{bulk}} = \arccos\left(\frac{\rho_{\text{bulk}}^{4k} - \eta_-^2 - \eta_+^2}{2\eta_- \eta_+}\right) , \quad (50)$$

$$\rho_{\text{cub}} = |\eta_- + e^{2\pi i/3} \eta_+|^{1/(2k)} . \quad (51)$$

Here ρ_{bulk}^{2k} is the target return modulus, so the numerator in (50) contains its square ρ_{bulk}^{4k} . These formulas use a scalar aligned-eigenvector model; the dense audit below tests that approximation at one feasible noise.

8.2 The bright return is nearly one-channel

Table 1: Two-profile branch compression. The compressed bright and full branch-complete columns are one-step radii. The final column is the modulus ratio of the two compressed eigenvalues.

σ	k	compressed bright	full two-branch	dark/bright
10^{-2}	3	0.798145	0.801218	4.07×10^{-2}
4×10^{-3}	4	0.794331	0.794838	2.00×10^{-2}
2×10^{-3}	5	0.801312	0.802324	2.70×10^{-4}
10^{-3}	6	0.791254	0.791673	4.01×10^{-5}
5×10^{-4}	6	0.777651	0.777588	1.35×10^{-2}
2×10^{-4}	7	0.784393	0.784378	3.99×10^{-3}
10^{-4}	8	0.789706	0.789825	1.45×10^{-5}

The two-profile bright radius follows the complete positive two-branch return to within 3.1×10^{-3} in absolute one-step radius at every scale and much more closely in the tail. This supports a low-dimensional description of the *bright branch return*. It does not identify that return with the physical bulk edge: both bright columns remain near 0.79 at the smallest noise, whereas the bulk edge is near 0.757.

At $\sigma = 10^{-3}$, the raw two-profile matrix is

$$B_{\text{raw}}^{(2)} = \begin{pmatrix} 0.03017623 & 0.03034803 \\ 0.02988284 & 0.03004815 \end{pmatrix}, \quad (52)$$

with eigenvalues

$$0.0602268018, \quad -2.41557 \times 10^{-6}. \quad (53)$$

In the bright/dark basis,

$$U^* B_{\text{raw}}^{(2)} U = \begin{pmatrix} 0.06022763 & -1.68555 \times 10^{-4} \\ 2.96636 \times 10^{-4} & -3.24571 \times 10^{-6} \end{pmatrix}. \quad (54)$$

The matrix is not exactly exchange symmetric, but its dominant channel is clearly bright.

After applying Perron/parity deflation before every localization,

$$B_{\text{b}}^{(2)} = \begin{pmatrix} 0.01889414 & 0.01908193 \\ 0.01871482 & 0.01889609 \end{pmatrix}, \quad (55)$$

with eigenvalues

$$0.0377925994, \quad -2.37049 \times 10^{-6}. \quad (56)$$

Deflation changes the bright amplitude by order one but does not reveal a hidden antisymmetric eigenvalue.

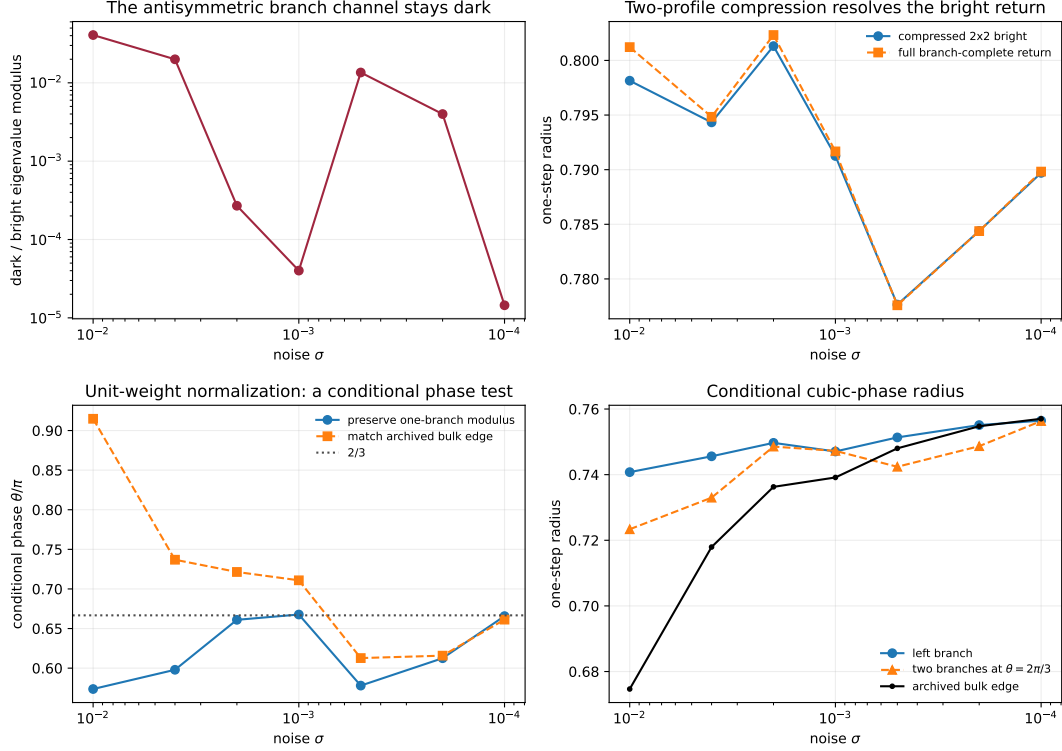


Figure 1: Seven-noise branch audit. Top left: the smaller two-profile eigenvalue remains dark relative to the bright return. Top right: the two-profile bright radius resolves the full positive two-branch return. Bottom left: phases inferred under the additional unit-weight assumption; the staircase is not monotone. Bottom right: the fixed cubic-phase radius, left-branch radius, and archived bulk edge. All comparisons are ordinary floating-point diagnostics.

8.3 Conditional phase diagnostics

Table 2: Scalar phase diagnostics from the separate left and right returns. θ_{one} preserves the left-branch modulus; θ_{bulk} matches the archived bulk radius.

σ	k	η_+/η_-	θ_{one}/π	θ_{bulk}/π	ρ_{cub}	ρ_{bulk}
10^{-2}	3	0.45858	0.57364	0.91484	0.723357	0.674669
4×10^{-3}	4	0.60571	0.59794	0.73689	0.732983	0.717941
2×10^{-3}	5	0.96904	0.66101	0.72144	0.748550	0.736278
10^{-3}	6	1.00565	0.66771	0.71090	0.747245	0.739142
5×10^{-4}	6	0.48488	0.57795	0.61271	0.742403	0.748015
2×10^{-4}	7	0.69242	0.61253	0.61578	0.748676	0.754749
10^{-4}	8	0.99476	0.66570	0.66127	0.756336	0.757023

The staircase changes in k also change the critical clearance δ_k/σ , so branch equivalence is not monotone across this short sequence. There is no seven-point convergence theorem in table 2. The relevant observation is narrower: at the two most branch-symmetric points, $\sigma = 10^{-3}$ and 10^{-4} , θ_{one}/π is 0.66771 and 0.66570, close to $2/3$.

At the smallest noise,

$$\frac{\theta_{\text{one}}}{\pi} = 0.6657039620, \quad \frac{\theta_{\text{bulk}}}{\pi} = 0.6612734223, \quad (57)$$

while

$$\rho_{\text{cub}} = 0.7563358794, \quad \rho_{\text{bulk}} = 0.7570230790. \quad (58)$$

The cubic-phase error is

$$\rho_{\text{cub}} - \rho_{\text{bulk}} = -6.8720 \times 10^{-4}. \quad (59)$$

This is a good numerical clue. It is not enough to choose between the two normalizations in [proposition 5.3](#).

9 Dense phase family at $\sigma = 10^{-3}$

The scalar formula uses only the principal eigenvalue of each positive branch return. At $\sigma = 10^{-3}$, the endpoint window has dimension 68, small enough to materialize the two 68×68 returns and compute the full family

$$R(\theta) = R_- + e^{i\theta} R_+, \quad 0 \leq \theta \leq \pi. \quad (60)$$

The same calculation is repeated after Perron/parity deflation is inserted before every packet restriction.

Table 3: Dense endpoint phase family at $\sigma = 10^{-3}$ and $k = 6$. s_2/s_1 is the second/first singular-value ratio of $R_- + R_+$. All radius columns are converted to one Markov step.

operator	s_2/s_1	$\rho(0)$	$\rho(2\pi/3)$	$\rho(\pi)$	θ_{bulk}/π
raw	0.002737	0.791673	0.747225	0.564358	0.710764
bulk-deflated	0.004223	0.756816	0.714420	0.583114	0.457379

The raw second singular value is only 0.274% of the first. This provides an endpoint-space check of the near-rank-one conclusion drawn from the two-profile matrix. The dense raw phase matching the bulk edge, 0.710764π , also agrees with the scalar estimate 0.710897π to $1.33 \times 10^{-4}\pi$.

The deflated family gives the more important warning. At $\theta = 0$ its radius is 0.756816, at the cubic phase it is 0.714420, and the crossing with the archived bulk edge 0.739142 occurs at 0.457379π . Deflation does not merely subtract a phase-independent scalar from the raw bright eigenvalue. A phase fitted before peripheral extraction is therefore not a physical sector identification.

9.1 Numerical status and reproducibility

The implementation unit-tests the forced-phase formula, cubic-phase rank-one law, bright/dark transform, Sylvester nonzero-spectrum identity, profile construction, compressed cycle, and dense matrix materialization. The full audit records all matrix entries, phase samples, software versions, source hashes, and inherited-data hashes. NumPy, SciPy, and Matplotlib provide the numerical implementation [2, 3, 7]. The largest grid has $n = 204800$ midpoint cells. No interval arithmetic, directed rounding, operator-norm enclosure, or non-normal pseudospectral bound is used.

10 The corrected next operator target

The two-channel calculation turns the RH-19 obstruction into a more precise choice. The corrected determinant cannot be obtained by adding both positive branches and then taking the principal root. It must derive how the two branch contributions enter after peripheral extraction.

A viable next construction has four parts.

- G1. Biorthogonal branch data.** Construct entrance and exit maps $R_+(z) : \mathbb{C}^2 \rightarrow \mathcal{H}$ and $R_-(z) : \mathcal{H} \rightarrow \mathbb{C}^2$ whose ranges and dual ranges are orthogonal to the Perron/parity right

and left modes. This may produce nontrivial real branch weights before any phase is introduced.

- G2. Derived closing multiplier.** Compute the spectral-parameter boundary map that closes the two critical branches. If it is unitary to leading order, test whether its relative eigenphase tends to $2\pi/3$. If it instead supplies a dual factor $1/2$, the cubic-phase interpretation should be discarded.
- G3. Uniform rank-one estimate.** Prove, in a balanced norm and along a specified noise subsequence or clock cell, that the branch effective Hamiltonian equals a rank-one bright block plus an error smaller than the required root-spacing scale. The finite ratios in [table 1](#) are evidence, not this theorem.
- G4. Sector-resolved complement bound.** Only after the first three steps should the remaining far-tail and bulk excursion spaces be inserted through a Schur self-energy. The contour must be chosen for the physical effective determinant, not for the replicated sector-free time lift.

The decisive calculation is G1–G2. It resolves the half-weight versus cubic-phase ambiguity before another large complement-resolvent estimate is attempted. Either result would be useful: a derived cubic phase would explain the observed cancellation mechanism, while a derived real normalization would remove an attractive but unnecessary phase hypothesis.

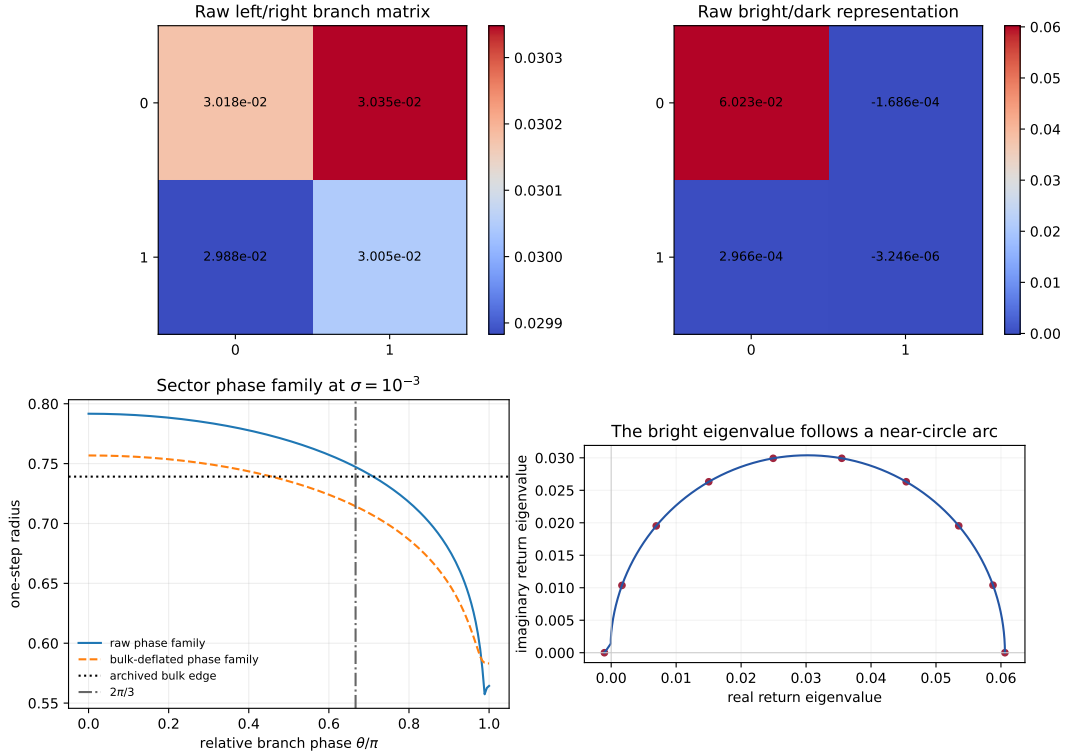


Figure 2: Dense phase-family audit at $\sigma = 10^{-3}$. Top: the raw two-profile matrix and its bright/dark representation. Bottom left: raw and bulk-deflated one-step radii as the relative branch phase varies; the dotted horizontal line is the archived bulk edge. Bottom right: the raw leading return eigenvalue follows a near-circular arc, as predicted by the rank-one formula. The switch near $\theta = \pi$ reflects competing small eigenvalues of a non-normal return.

11 Discussion

11.1 What has advanced

The two critical branches are no longer treated as an uncontrolled scalar complement. They form an exact direct-sum channel, and the endpoint return can be moved to that channel without changing its nonzero determinant. In the numerically dominant rank-one sector, the factor-of-two positive return, the dark antisymmetric mode, and the effect of a relative phase are all described by one scalar formula.

The cubic phase is therefore not an arbitrary curve fit. It is the unique unit-modulus relative phase that preserves one-branch amplitude in the exchange-symmetric rank-one model. The smallest-noise data approach both the required branch symmetry and the predicted angle. This is enough to justify testing the mechanism in a derived effective Hamiltonian.

11.2 What has not advanced

The present calculation does not identify the branch phase of the physical Gaussian operator. It does not prove full-return rank-one convergence, a uniform complement resolvent estimate, or a one-to-one correspondence between local roots and distinct physical eigenvalues. The dense deflated benchmark shows why those statements cannot be inferred from the raw phase family.

The non-identifiability proposition is equally important. A half-weighted positive sum reproduces the same symmetric leading radius, so radial agreement cannot establish a complex phase. Angular eigenvalue data, biorthogonal entrance/exit maps, or an exact determinant combinatorics are needed.

11.3 No arithmetic or self-adjoint implication

Every operator in this paper comes from one noisy quadratic map. The cubic root of unity is a conditional branch-closing coefficient, not a zeta zero. No result constructs a self-adjoint Hilbert–Pólya operator, proves a prime-power trace formula, identifies the Riemann zeros, or implies the Riemann hypothesis. Any future arithmetic comparison would require an independent trace identity not present here.

12 Conclusion

The branch-resolved correction survives the RH-19 no-go test in a precise but conditional form. Endpoint and branch returns are linked by an exact Sylvester factorization, so the corrected two-channel determinant is canonical. A one-profile rank-one reduction has a bright symmetric mode and a dark antisymmetric mode; its phase-dependent eigenvalue is $\eta_- + e^{i\theta}\eta_+$.

If the branch weights are constrained to have unit modulus, exact branch equivalence forces $\theta = \pm 2\pi/3$ when the combined return is required to retain one-branch modulus. At $\sigma = 10^{-4}$, both the independently inferred phase and the archived bulk radius lie close to that conditional prediction. The two-profile dark/bright ratio is 1.45×10^{-5} .

The result is not yet a physical phase theorem. A real half weight gives the same symmetric radial correction, discrete time sectors do not contain the cubic phase for general k , and peripheral deflation substantially changes the dense phase family. The next gate is therefore well defined: derive the two branch weights from a peripherally biorthogonal, spectral-parameter Grushin problem. That derivation, rather than another radial fit, will decide whether the cubic phase is a genuine mechanism or a useful but ultimately unnecessary coordinate choice.

Data and code availability

The source code, tests, complete CSV and JSON audits, figures, and manuscript are available with this paper [13].

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